

Simulation the Behavior of Robot Sub-Swarm In Spatial Corridors

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Abstract – The article presents simulation of the robot swarm behavior, taking into consideration the model of robots dividing into sub-swarms to perform similar tasks within a limited spatial corridor. To avoid robots crashing with each other, the authors propose to use the three-dimensional (3D) bodies (sphere) instead of spatial points in the traveling salesman problem (TSP) solution. For simplifying the solution of such problem, it is proposed to solve 3D problem via concatenation of solving two 2D problems of moving sphere, as “salesmen” move in planes, limiting the spatial corridors. The sphere in this case moves on the vertical plane notwithstanding the spatial corridors. The proposed simulation model of the swarm robots behavior is described as dynamic smart system.

Keywords – robot sub-swarm, UAV, UGV, spatial corridor, Traveling Salesman Problem (TSP), 2D&3D simulation.

I. INTRODUCTION

The autonomous robot swarms are used as unmanned ground vehicle (UGV), unmanned underwater vehicles (UUV), unmanned water (surface) vehicles (USV) or unmanned aerial vehicles (UAV aka: “drones”). Production tasks using autonomous robot swarm provide algorithmic movement of robots within a limited space. It is rather difficult to make a robot or several of them appear in a predefined point with clearly indicated coordinates. Moreover, each point has to be visited by at least one of the robots.

Mathematically, this problem is reduced to the classic traveling salesman problem (TSP) [1]. The actual problem of robot behavior as a “traveling salesman” is described not only in geographical coordinates (2D-problem), but also in terms of (3D), where a robot fulfills a task [2].

In addition, the swarm of robots used in the manufacture is often divided into groups (sub-swarms), which perform their own tasks. Though different groups perform similar tasks, each group acts in its spatial corridor.

The problem is that the existing algorithms for TSP solutions in searching for the optimum route spend a significant amount of computing resources and time – (more than 10 sec to calculate 20 waypoints [3]). However, in a fast dynamic process where the UAV speed is from 5 to 15 m/s [4], the spatial corridor can be passed faster if the best route is calculated.

The above said leads us to the possibility of transforming a 3D TSP into a 2D task. Such 2D TSPs can be paralleling

solved on different computing resources, and followed by concatenation of the found solutions.

In classical solutions of 3D-spatial tasks a “salesman” is a point without any physical volume [5, 6]. In real-time production environment, either this approach could lead to collapse of the individual or several robots when they crash with each other, or it could require additional calculations to avoid collision with other robots of the swarm. Both outcomes need preliminary mathematical simulation to prevent critical situations while using the robot swarm in a spatial corridor.

The article deals with modeling the group behavior of robot swarm via dividing them into sub-swarms to perform similar tasks in the limited spatial corridors.

II. BACKGROUND

A. Task types that use spatial corridors

One of rather complex tasks is the implementation of computer/robotic technologies into pharmaceutical warehouse. According to Ukrainian laws, the pharmacy placement area should be at least 300 square meters, perhaps on two floors [7].

Then to support such equipment it is expediently to install robots (Robowarehouse) Rowa Vmax/ Rowa Smart (6 units in 3 spatial corridors) as in Fig. 1 [8]. The capacity of Rowa robots (UGV) is up to 4000 packages per linear meter. Omitting the simulation phase before starting physical robotic pharmaceutical warehouse is too risky in the real world, as it can lead to damaging or even collapsing the whole swarm (robotic system) in automated pharmaceutical warehouse. There can also be significant losses in medicine stocks.

In this case, it is necessary to optimize the route of robot sub-swarms which operate in the same space: 6-FS1 & 6-FS2, 6-FS3 & 6-FS4, 6-FS5 & 6-FS6 (on Fig. 1, a, in the room No. 6).

A similar problem of the robot sub-swarm motion in a spatial corridor can be seen in drones movement while constructing a rope bridge (on Fig. 1, b). It should be noted, that it isn't a classical TSP statement. To build a durable construction, withstanding a man, you should compose the rope bridge of different elements, such as knots, links, and braid [9]. That is why a UAV-“salesman” has to go through some route nodes several times.

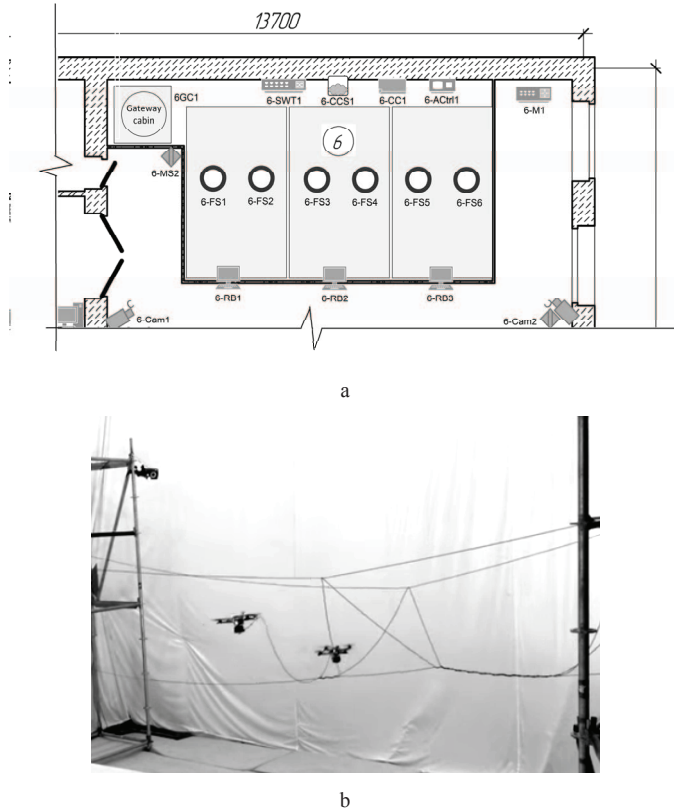


Fig. 1. Location of robot swarm (FS) in the pharmaceutical warehouse (a) and in the spatial corridor (b)

B. Description of the mathematical model

Considering the fact that in both cases on Fig.1, there is a corridor with cells, we do not need (when moving from one side of the corridor to another) to count the inside points of the spatial corridor. Therefore, such a problem can be considered not as a 3D task, but as a composition of two 2D TSP tasks: on the left vertical plane and right vertical plane in this corridor separately.

We consider the agreed robot motion to solve a common problem for the whole swarm selection of the medicine, etc. there If there is no single control center, unmanned vehicles are should regulate their actions on one's own and determine their routs considering motion of each robot neighbor. To better address, the tasks of coordinated movements it is necessary to highlight the procedure of planning, where each swarm robot determines its initial steps.

Here, we can do without complex formulas and focus on mathematical model of objects interpretation and understanding this model.

The algorithms for solving this TSP are based on the search of the shortest possible distance on the graph from the starting point to any desired point as the end of the route.

On the other hand, the known algorithms do not take into account appearing obstacles, such as robot swarm members, which can change their position eventually. This leads to the fact that the time of gaining the target is increased.

To start the task we need:

- a list of names medicines / rope knots / other units and their number;
- location coordinates of such units on the sides of the plane spatial corridor (in the cell);
- each robot's own coordinates.

In the process, these data are subject to change:

- the list of necessary medicines / other units must be reduced;
- the robot coordinates are changing;
- the coordinates of all robot swarm members are changing too.

It should be noted, if the delivery robots are used (as in pharmaceutical warehouse), they should return to the start position for unloading medicines out and for laying them into boxes.

Since the exchange of information within the swarm is limited, the use of path search algorithms (such as Dijkstra's algorithm, Lee algorithm, etc.) will not be effective in the process of moving to the goal.

The most suitable option would be to plan concerted action of team members, and then follow this plan. At the beginning, all of the team must make an initial choice of place to be able to assess its performance of other selection option purposes. Each robot determines the closest target.

The choice of targets (medicine) is performed by means of a special matrix choice:

$$T = || t_{ij} ||, (i=0,1,...,n; j=1,2,...,m). \quad (1)$$

Matrix selection (1) is a matrix of size $n \times m$, where n is the total quantity of robots; m is the number of medicine items of, X_{nm} is the distance to reach for cells with medicine m . Each robot calculates the minimum value of $X_{mn} \min$ and finds it the most optimal. Then the distance is checked and the robots, which has the smallest distance and no other problems, takes the task to perform. After these operations, the choice of targets is made again. If the targets are not changed, the chosen plan is adopted.

Thus, appears an opportunity to assess the real time for which a robot achieves goals without considering all possible obstacles that may arise on its way. This scheme allows you to really select the optimum plan.

To implement the accountability of environmental dynamics (the ability to track the movement of other robots) you should receive new data after a certain time period and possible route overlapping's, before setting a hierarchy of robots' stopping.

The aim of this work is also to create a dynamic IP prototyping system that includes component of movement simulation modeling and execution of production activities (collecting medicines, tying knots etc.) automatic devices

in a confined space with moving obstacles. The following set-theoretic model represents a simulation model of the system:

$$Msm = \langle MMO, MC, MIE, VU, VE, VY, VZ, S, F1, F2 \rangle (2),$$

where *MMO* – a model of the managed object (*MO*). It describes the condition and characteristics of the robot;

MC – a model of controller. It describes the principles of robot operations;

MIE – a model of internal events. It describes the cases of random events and their characteristics (medicines' weight, medicines' number of etc.);

VU – lots of controlled and managed parameters: dynamic (mobile) obstacles encountered on the robot's way, robot's work characteristics (speed of movement, gathering speed, maximum charge of battery, medicines' weight etc.);

VE – lots of random disturbances, which suddenly arise as dynamic obstacles, moving on a route given randomly;

VY – lots of output parameters used in the controller: coordinates of a robot, its status and environment;

VZ – lots of output parameters *MMO* robot's behavior in given conditions, a complete list of collection;

S – lots of system's modes: all possible combinations of elements of sets *VU*, *VE*, *VY*, *VZ*;

F1 – a function of generating a managing vector: the rules and procedures responsible for establishing new values of managed parameters;

F2 – a function displaying of *MO*'s input to its output: rules and procedures by which the output parameters value is calculated.

Let us consider the motion control. In frames of *VY* model management tasks, the motion control is in charge of robots' twists and vehicle movements, etc. It is necessary to calculate the position and orientation of a solid body at discrete points in time. The transferred control command includes a code for each of the predetermined maneuver parameters and specified maneuver data from maneuver beginning to its completion. Objects carry out complex spatial movement, which is viewed as a combination of translational and rotational motions of a point or a set of points.

In this paper, the approach to the implementation of the rotation algorithms is associated with the use of integer arithmetic. Among the algorithms that can solve this problem, we chose Braccini algorithms [10], which are actively used by foreign researchers and engineers for hardware implementation turns.

Braccini algorithm for generating points on a straight line [10]. Let us consider an algorithm which implements linear motion using only integer operands. According to Braccini, a point with integer coordinates (*X*, *Y*) belongs to a straight line emanating from the origin with a slope $k = (\Delta Y / \Delta X)$, if the following condition is performed:

$$|Y - (\Delta Y / \Delta X) * X| \leq \varepsilon, \quad (3)$$

where $|\varepsilon| \leq 0,5$.

Using only integer variables *X*, *Y*, ΔX , ΔY , we obtain the condition for a point:

$$-\Delta X \leq 2 * \Delta Y * X - 2 * \Delta X * Y \leq \Delta X. \quad (4)$$

If *X* values grow in sequence, and differ by one, the corresponding *Y* values can be found on the basis of this relation verification.

With reference to robots management task, the algorithm can be used to calculate the route of the object at any given angle in real time.

Operation of rotation based Braccini algorithm [10]. Let us consider the problem of a plane turn. The source object (a body) is generally represented in the form of matrix elements – a system of physical points. Moreover, we have a single point (*X*, *Y*), when consider the simplest case. It is necessary to carry out a system of points rotating for a predetermined angle.

The proposed algorithm is based on the rotation using Braccini model [10]. The new coordinates (*X**, *Y**) of the point (*X*, *Y*) are defined as follows:

$$X^* = (M/L) * X + (N/L) * Y$$

$$Y^* = -(N/L) * X + (M/L) * Y. \quad (5)$$

When applying a rotation conversion, the rotation angle α is generally given by the triplet (*M*, *N*, *L*) of integers, and $L^2 = M^2 + N^2$, $M/L = \cos \alpha$, $N/L = \sin \alpha$.

Sine and cosine table as a fraction of the integers is calculated in advance and stored in the memory of the turning system control.

A sequence of *X* and *Y* coordinate values may be checked with the help of *X** and *Y** corresponding values on the basis of this relation verification:

$$|2LX^* - 2MX - 2NY| \leq L$$

$$|2LY^* + 2NX - 2MY| \leq L. \quad (6)$$

The time of rotation algorithm execution is estimated as $\theta(k)$, where *k* – number of object points.

The algorithm uses only integers or its implementation. It is not based on the direct calculation of trigonometric functions, which ensures its application feasibility in systems of object management, and in mobile moving systems with computing resources restrictions. It is assumed that the integer algorithms form the basis of full-function controls of the route motion devices with limited computing capabilities to perform tasks. Moreover, built-in on-board computer software will provide orientation angles and motion parameters definition; navigation and control when moving along a predetermined path; selection and orientation of spatial corridor angles.

The described below data model was made for using this set-theoretical model properly. Every production unit here has its own placement parameters such a number of shelving, a shelf number, and a special position code. The data model was built in a software environment ARIS Platform (Fig. 2).

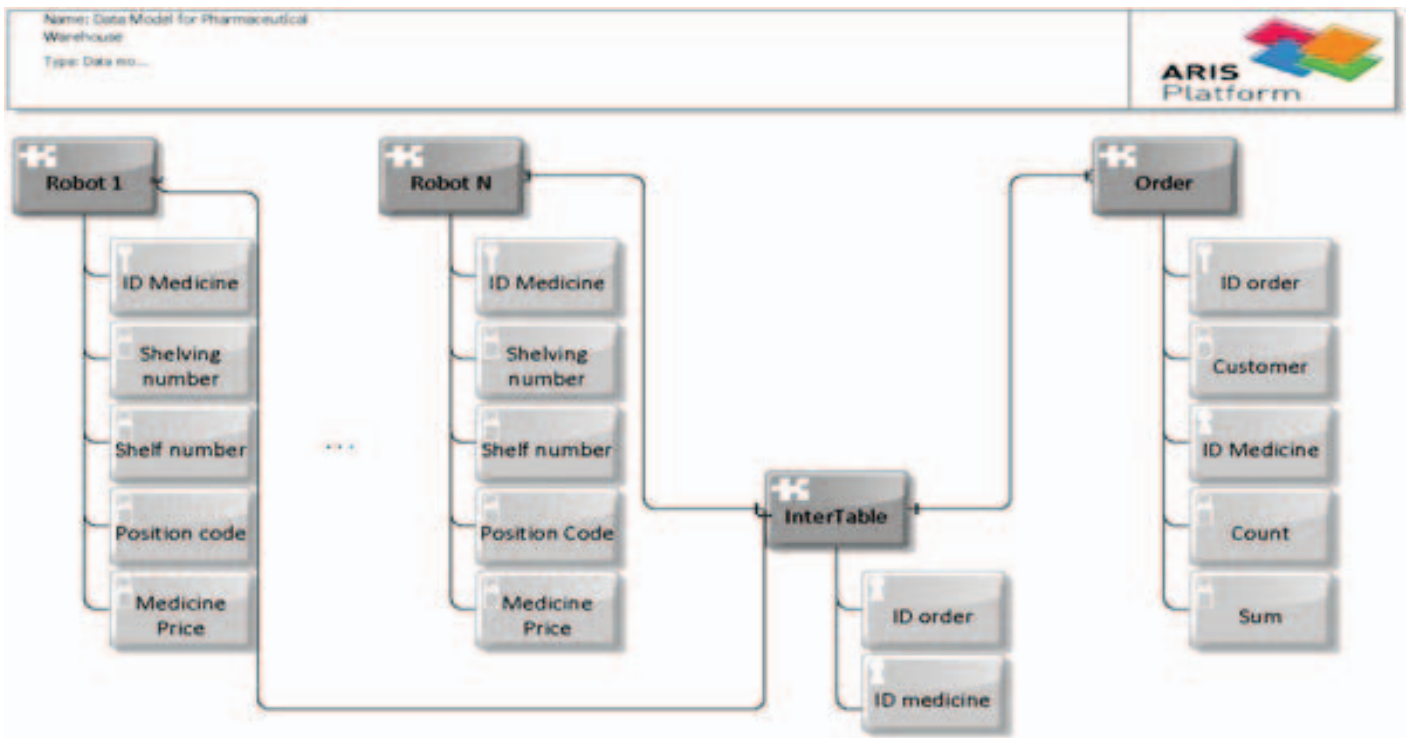


Fig. 2. Data Model for Pharmaceutical Warehouse

III. ROBOTS' BEHAVIOR SIMULATION

Special attention should be paid to the solution of problems similar to the TSP in the separate space corridors. At this rate it is impossible to calculate the steady mode of a robots' network by successive approximation error calculation. All the coefficients of the matrix (1) should be calculated in a single cycle, after which the overall objective for swarm robots is divided into sub-tasks (for each of its sub-swarm). The network gets to a stable mode at this point, and it remains in it. It means that after distribution sub-tasks within the robot sub-swarms, these tasks will not be redistributed until their implementation.

This algorithm formulation is the closest to the Hopfield network learning algorithm [11] where a sign of finding a solution is such a stable mode which coincides with one of the stored mode at training. Taking into consideration the fact that medicine location on the shelves does not change, this particular medicine will find medicine in the same place without errors next time.

It makes sense to consider a simultaneous mode of Hopfield network functioning. When paired robots of the sub-swarm have complied the task of scanning and selecting the medicines each on its own side of spatial corridor, their modes are changed to new ones simultaneously. As a result, they can exchange the corridor sides with each other. Otherwise, they can encounter and collapse the network functioning. Thus, it is necessary to simulate parallel operation of the robot sub-swarms applying the serial algorithm.

The work of the pharmaceutical/building robot sub-swarms, each moving in its own spatial corridors was simulated.

Each robot consists of two mechanisms and moves in its own aisle (Fig. 3), because, it is necessary to simulate a collision detect and a collision avoidance during the movement of each robot sub-swarm separately. All sub-swarms have different moving directions.

The productivity of order processing increases by dividing the task between two such robots. That's why the speed of the order making increases.

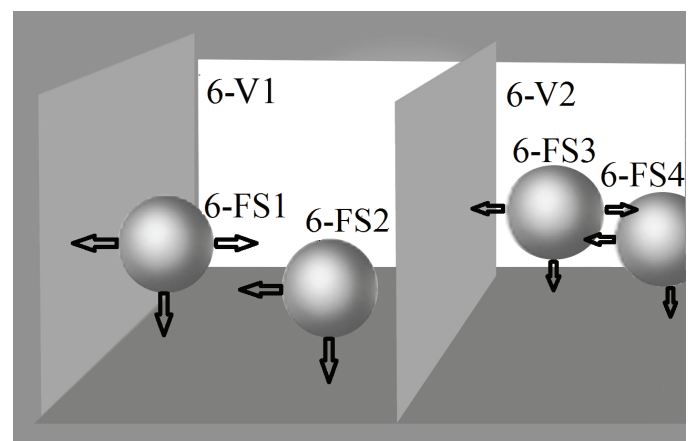


Fig. 3. Paired robots' movement

One robot performs tasks not only on the nearby shelves but also on the others due to its ability of 180 degrees rotation. The enumerated advantages increase the performance and improve the request processing speed. The ultimate way is to coordinate the issuance to the window.

Fig.4 shows the simulation of a solution finding process by using Hopfield networks [12]. In some situations, several mechanisms moving one after the other in the aisle can be used inside one robot.

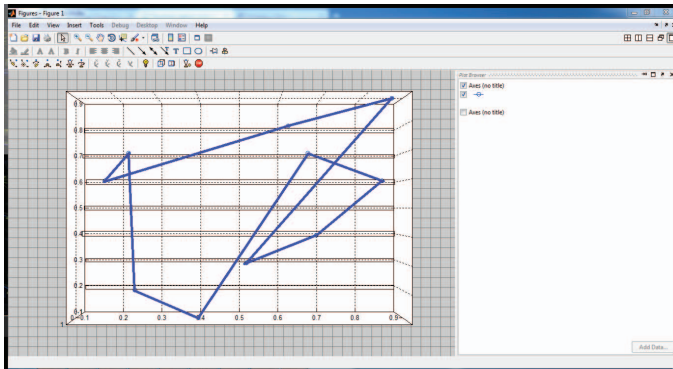


Fig. 4. Example of using Hopfield networks in TSP

Fig. 4 shows that Robot 1 (6-FS1) performed the 2D TSP on one side of the spatial corridor 6-V1 up to 0.5 of abscissa axis. At the same time, Robot 2 (6-FS2) performed a similar task on the other side of this corridor. After 0.5 value, Robot 1 changed the corridor side and continued to perform its 2D TSP sub-task on the vertical plane of the second side of the spatial corridor 6-V1.

Robots used in the above example aren't independent but they are paired as one mechanism. So, it is necessary to implement Hopfield network with parallel computing to build a similar Fig. 4 route for Robot 2. In this case, to calculate Robot 2 position, you need to know an updated position of Robot 1. It is clear that it is physically impossible to calculate both positions simultaneously, because Robot 2 position depends on the position of Robot 1.

If you increase a number of robots in a sub-swarm for calculating a new position for each robot, you should know the old and updated position of neighboring robots. The number of clock cycles to transfer robots' positions increases with a number of neighboring robots whose position should be taken into account.

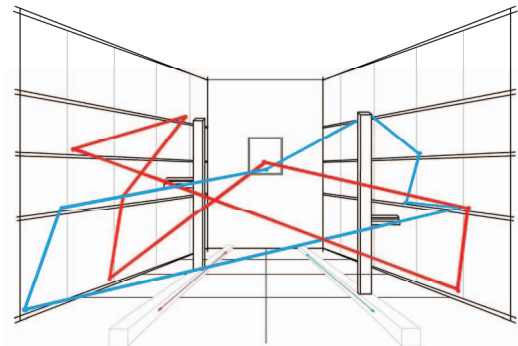
The entire above said let us talk about implementing Hopfield network with an asynchronous mode.

Because two robots can handle two racks efficiently, quickly and without collisions, it is appropriate to introduce each of them in the form of spheres with a set diameter in the simulation (Fig. 5, b, c).

While solving the problem of robot motion simulation, three spatial coordinates have been considered:

- the 1st coordinate: Left or right rack;
- the 2nd coordinate: Position on the shelf;
- the 3rd coordinate: The actual number of shelves.

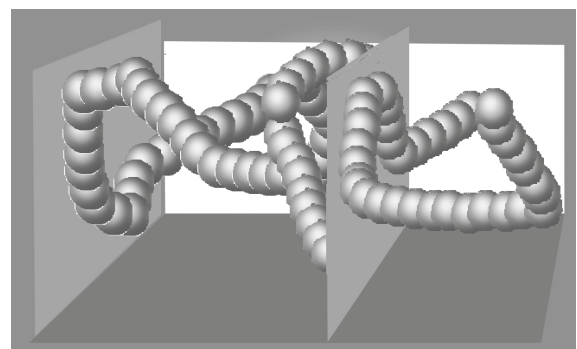
Simulation of robot sub-swarm behavior can be used in a preparation phase for the order implementation in automated pharmaceutical warehouse as well as in optimization of robots sub-swarm cooperative motion in limited space between the shelves of pharmaceutical warehouse.



a



b



c

Fig. 5. Robot motion simulation in the spatial corridor: (a) "salesman" as spatial point, (b) "salesman" as circle, (c) sub-swarms as spheres' route in spatial corridors

Thus, the simulation is very important in the study of robots' swarm behavior; because it gives an opportunity to present a real system, using programmed mathematical models, and allows to display the dynamics of the system as well as to investigate the behavior of segmented way.

It should be noted, that implementation of this approach does not mean that sub-swarm path will be minimal because the described solution is only sub-optimal for TSP. However, Hopfield network usage allows finding satisfactory solutions quickly without further analysis. This makes usage of the proposed method possible as it develops the formation

of robot sub-swarm route via a concatenation of several 2D solutions in 3D space in speed dynamical networks, such as a UAV swarm. Application of classical methods with multiple iterations to find an optimal solution of TSP may require such a great time and computing resources that will collapse the UAV swarm functioning when it receives a route correction late.

IV. CONCLUSIONS

In this article, we presented the simulation of robot swarm motion, when all robots are divided into several sub-swarms spatially. Each sub-swarm works in its own spatial corridor, but all robots perform similar tasks.

Classic TSP calls for finding the solution in 2D space, but during the detailed research of the subject, we found out that a robot could move in three directions. To decide the TSP as a 3D task correctly, it is suggested to percept the robot as a sphere instead of a point.

Taking into account the fact that both delivering, and building robots perform tasks only on the vertical planes of the corridor, it is proposed to solve the 3D TSP for left and right vertical planes separately, Author suggest to end this process by concatenating two 2D TSP in the general solution for robot swarms.

The proposed simulation model of the robot swarm behavior is described as a dynamic smart system.

Simulation process of two robots management that represent a common mechanism (i.e., moving synchronously, but performing different tasks) is presented in this paper.

Note, however, that these algorithms can't guarantee finding the most optimal route. Their purpose is to shorten the time of finding an acceptable solution to this problem. That is, these algorithms will optimize your decision as much as you have time for it.

Our research work is currently in progress, and the main objective is to improve the proposed algorithm in order to create brand new software, that can help to simulate the process of the optimal route creation.

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